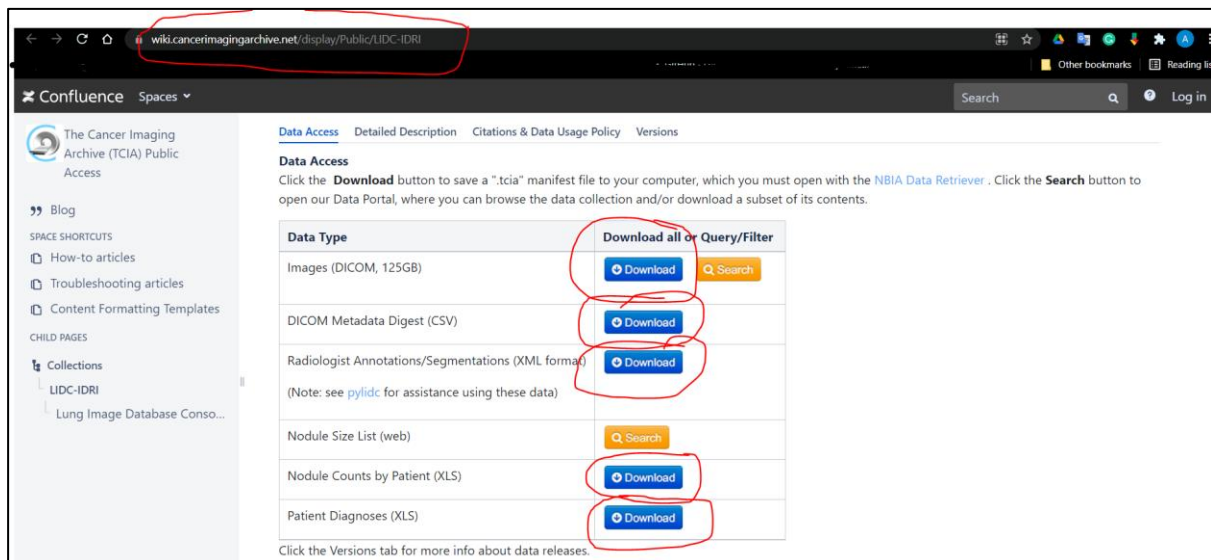


STEPS TO BE FOLLOWED FOR EXECUTION OF THE PROJECT

1. DOWNLOAD THE DATASET
2. SET WORKING DIRECTORY AND WORKING ENVIRONMENT IN ANACONDA PROMPT.
3. INSTALL NECESSARY LIBRARIES IN THE ENVIRONMENT
4. RUN JUPYTER NOTEBOOK & OPEN NOTEBOOK FOR “CANCER_DETECTION”
5. EXECUTE EACH CELL ONE-BY-ONE FROM THE TOP IN THE NOTEBOOK.
6. THE LAST STEPS ARE FINAL RESULTS OF THE MODELS’ PERFORMANCE.

DOWNLOAD THE DATASET

1. Go to the link : <https://wiki.cancerimagingarchive.net/display/Public/LIDC-IDRI>
2. Click on download for all the files, and store all these files in the special directory of dataset like “D:/Project_CancerDetection /Dataset”



3. On clicking the download button for Images (DICOM, 125 GB), a file with name “TCIA_LIDC-IDRI_20200921.tcia” will be downloaded. Open that file, “NBIA Data Retriever” will open, choose the option of “Classic Directory Name”, change the location of the download to folder, and start the download. This step may take some time depending on the internet connectivity speed. It is suggested that keep this processing running (data download) overnight.

SET WORKING DIRECTORY AND WORKING ENVIRONMENT IN ANACONDA PROMPT.

1. Open Anaconda Prompt terminal from start menu.
2. Change the working directory by :
 - a. Type the drive letter where all the data and codes are located
for example, type ' `D:` ' if all your codes and data are in D-Drive.
 - b. Change the working directory to the folder where the "Cancer_Detection.ipynb" file is located, using the command ' `cd PATH_TO_YOUR_FOLDER_CONTATING_CODES` ' for example ' `cd D:/Project_CancerDetection/Codes` '
3. Change the working environment to the once that was originally created for this project
<please follow the previous guide if the working environment is not created>
Make sure that you have created the environment using 'Python=3.6'
type the command ' `conda deactivate` ' followed by the command
' `conda activate YOUR_ENVIRONMENT_NAME` '
for example: " `conda activate cancerDetection` "

INSTALL NECESSARY LIBRARIES IN THE ENVIRONMENT

1. Install the python libraries using the command
`'pip install LIBRARY_NAME==LIBRARY.VERSION '`

for example, `'pip install numpy== 1.19.2 '`

Install all the following libraries using the above mentioned command

PS. You might be asked to confirm the installation, accept the installation by entering 'Y'
followed by 'enter-button'

- a. `numpy== 1.19.2`
- b. `pandas== 1.1.3`
- c. `matplotlib==3.3.4`
- d. `dicom==0.9.9.post1`
- e. `scipy== 1.5.2`
- f. `skimage== 0.17.2`
- g. `tensorflow== 1.14.0`
- h. `Keras==2.3.0`
- i. `h5py==2.10.0`
- j. `notebook==6.4.0`

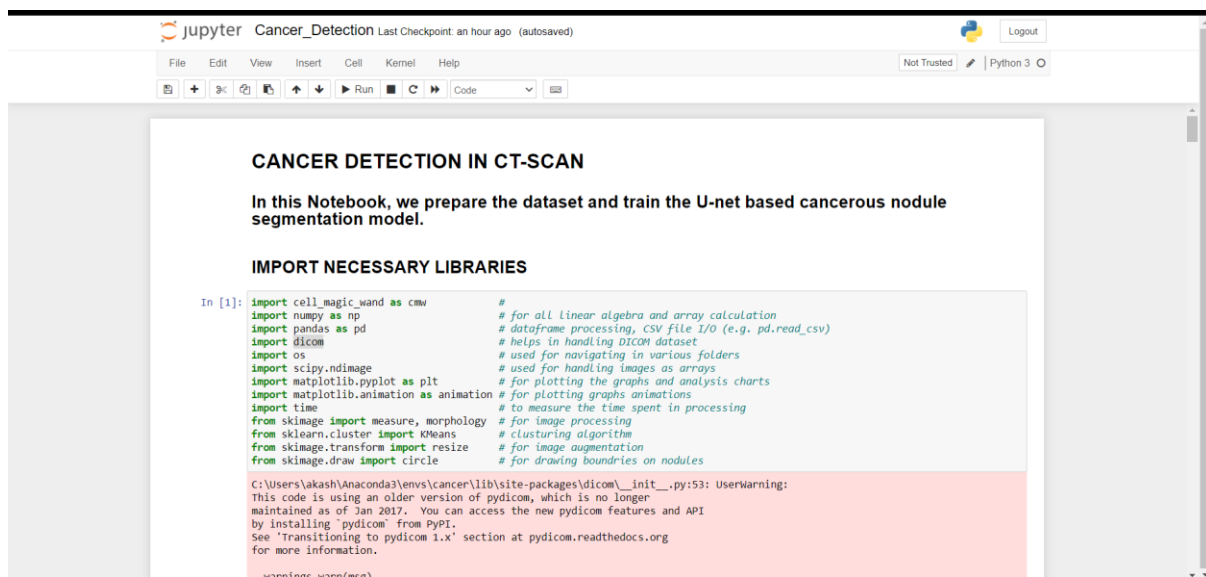
RUN JUPYTER NOTEBOOK & OPEN NOTEBOOK FOR “CANCER_DETECTION”

To run the Jupyter notebook, use the following command, in the anaconda prompt after setting up the working directory and the working environment

Command : ‘`Jupyter notebook`’

A new window will open in your browser, choose to open the “Cancer_Detection.ipynb” file.

It will look something like this :

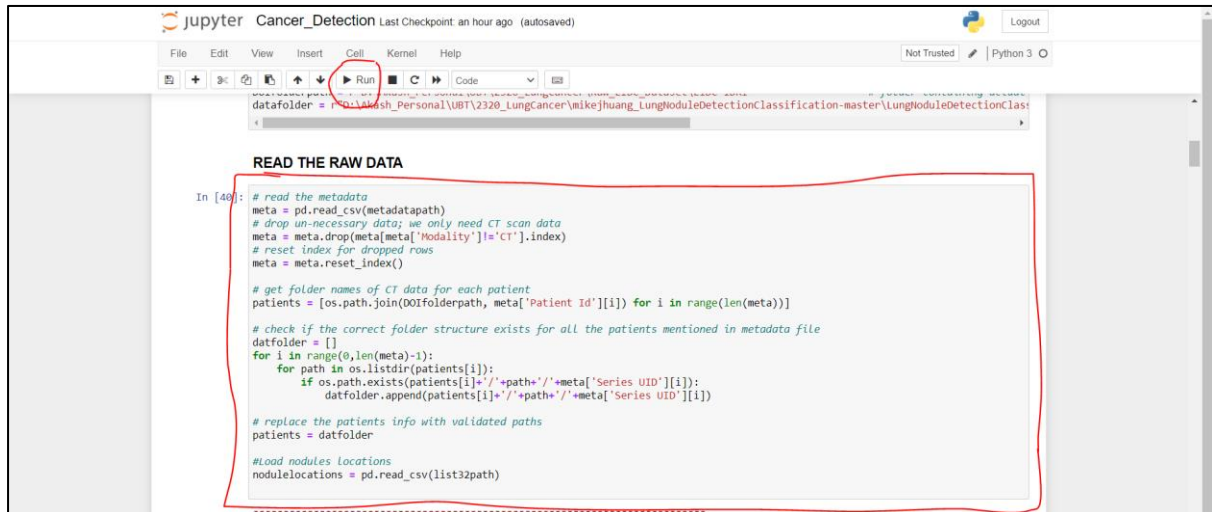


EXECUTE EACH CELL ONE-BY-ONE FROM THE TOP IN THE NOTEBOOK

Jupyter notebook uses the concept called cells, to define the codes.

One can write the code in cells, and execute the cell to check the working of the code.

To execute a cell, select the cell by clicking on it, and then click the run button at the top,



You need to change the paths in the cell : “SET NECESSARY PATHS”

Please copy the complete path to the following files:

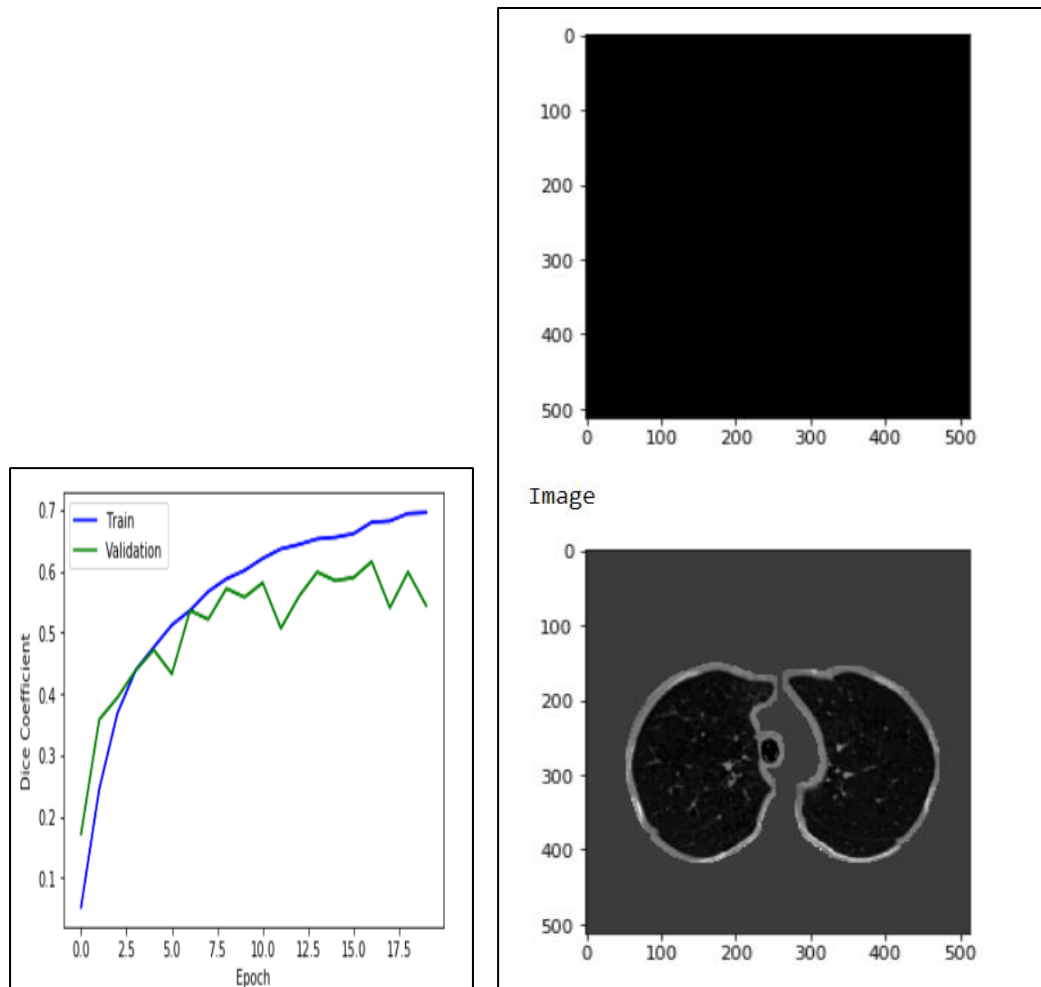
1. Metadatapath = Path to the “LIDC_IDRI_MetaData.csv” file.
2. list32path = Path to “list3.2.csv” file.
3. DOIfolderpath = Path of the folder where all your images dataset is stored
for example, if the dataset is present at “D:/Project_CancerDetection/Dataset/LIDC_IDR”
and this folder has the actual dataset folder, like LIDC-IDRI-001, LIDC-IDRI-002, etc etc. then
you need to copy the folder path as r“D:/Project_CancerDetection/Dataset/LIDC_IDR”.
Mine looks like :

Name	Date modified	Type	Size
LIDC-IDRI-0001	26-05-2021 08:53 AM	File folder	
LIDC-IDRI-0002	26-05-2021 08:54 AM	File folder	
LIDC-IDRI-0003	26-05-2021 08:54 AM	File folder	
LIDC-IDRI-0004	26-05-2021 08:55 AM	File folder	
LIDC-IDRI-0005	26-05-2021 08:56 AM	File folder	
LIDC-IDRI-0006	26-05-2021 08:56 AM	File folder	
LIDC-IDRI-0007	26-05-2021 08:57 AM	File folder	
LIDC-IDRI-0008	26-05-2021 08:57 AM	File folder	
LIDC-IDRI-0009	26-05-2021 08:58 AM	File folder	
LIDC-IDRI-0010	26-05-2021 08:58 AM	File folder	
LIDC-IDRI-0011	26-05-2021 08:59 AM	File folder	
LIDC-IDRI-0012	26-05-2021 09:00 AM	File folder	
LIDC-IDRI-0013	26-05-2021 09:01 AM	File folder	
LIDC-IDRI-0014	26-05-2021 09:01 AM	File folder	
LIDC-IDRI-0015	26-05-2021 09:02 AM	File folder	
LIDC-IDRI-0016	26-05-2021 09:02 AM	File folder	
LIDC-IDRI-0017	26-05-2021 09:03 AM	File folder	
LIDC-IDRI-0018	26-05-2021 09:04 AM	File folder	
LIDC-IDRI-0019	26-05-2021 09:04 AM	File folder	
LIDC-IDRI-0020	26-05-2021 09:05 AM	File folder	

4. Datafolder = path to a temporary folder, where all your temporary files will be saved.

THE LAST STEPS ARE FINAL RESULTS OF THE MODELS' PERFORMANCE

Once all the cells are executed one after another, the final (last) cells will display the results like



These are the final output from you AI model.

Please note that some cells can take a lot of time to complete execution, for example

1. The last cell before "MODEL TRAINING"
This cell has the code for preparing the dataset for training. It takes time depending on the computer hardware capabilities. This step can consume upto 6-12 hours also.
2. The cell for "TRAIN THE MODEL" will also take a lot of time depending on the hardware
This step can consume 4-5 days, as this step is the actual deep-learning model training step.

It is highly recommended that you use a computer with higher RAM size.